

Food Inventory System for IUT Cafeteria

CSE 4408

Group : 10% SAD

Lab - 09

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Project Context

An automated inventory system is required to track daily usage and manage stock levels efficiently

Briefly

- Manage inventory digitally
- Automate stock tracking
- Generate Financial and Stock alert



Data Flow Description

Name: Low Stock Alert Notification

Description: Carries alert details when an item's quantity falls below its set threshold

Source	Destination
Process – Notify Low Stock	• Inventory Staff • Cafeteria Manager • Notification Logs (D3)
Types Of Data Flow : ☐ File ☒ Screen ☐ Form ☐ Report ☒ Internal	
Data Structure : Low Stock Alert Record	Volume/Time : 3-5 / month
Comments • Alerts are generated in real time whenever the stock check process detects a shortage. • Considered high-priority data, requiring immediate attention from relevant personnel.	

Defined Data Structure

Data Structure Name : Low Stock Alert Record

Description :

Represents the complete data packet generated when an inventory item's current quantity drops below the predefined threshold.

Algebraic Notation :

Low Stock Alert Record = Item ID + Item Name + Current Quantity + Threshold Quantity
+ Alert Date + Alert Message

This structure supports both real-time alert display on the inventory dashboard and archival in notification logs for auditing and trend analysis.

Element Description Form

ID:

NAME: ITEM ID

ALIAS: ITEM CODE

DESCRIPTION

UNIQUE IDENTIFIER FOR EACH INVENTORY ITEM

LENTGH 6

INPUT FORMAT AA9999

OUTPUT FORMAT AA9999

DEFAULT VALUE NONE

☐ Alphabetic

☒ Alphanumeric

☐ Numeric

☐ Date

☐ Continuous or ☒ Discrete

☒ Base or

☐ Derived

VALIDATION CRITERIA

DISCRETE VALUE MEANING

ENSURES EACH ITEM HAS A UNIQUE REFERENCE

COMMENTS

- USED AS THE PRIMARY KEY FOR INVENTORY ITEMS.
- ENSURES ALERT LINKS DIRECTLY TO THE CORRECT ITEM RECORD.

Element Description Form

ID: _____

NAME: CURRENT QUANTITY

ALIAS: STOCK LEVEL

DESCRIPTION

QUANTITY OF ITEM CURRENTLY AVAILABLE IN STOCK

LENTGH 5

INPUT FORMAT 9999

OUTPUT FORMAT 9999

DEFAULT VALUE NONE

☐ Alphabetic

☐ Alphanumeric

☒ Numeric

☐ Date

☐ Continuous or ☒ Discrete ☒ Base or ☐ Derived

VALIDATION CRITERIA

DISCRETE VALUE MEANING

MUST EXIST IN IUT DATABASE'S ITEM LIST

COMMENTS

- USED FOR HUMAN-READABLE IDENTIFICATION IN ALERTS.
- PULLED AUTOMATICALLY FROM THE IUT INVENTORY DATABASE TO AVOID MANUAL ENTRY ERRORS.

Element Description Form

ID: _____

NAME: THRESHOLD QUANTITY

ALIAS: MINIMUM QUANTITY

DESCRIPTION

MINIMUM ACCEPTABLE QUANTITY BEFORE TRIGGERING ALERT

LENTGH 6

INPUT FORMAT AA9999

OUTPUT FORMAT AA9999

DEFAULT VALUE NONE

☐ Alphabetic

☒ Alphanumeric

☒ Numeric

☐ Date

☐ Continuous or

☒ Discrete

☒ Base or

☐ Derived

VALIDATION CRITERIA

DISCRETE VALUE MEANING

DEFINES SAFE STOCK LIMIT BEFORE ALERT

COMMENTS

- CRITICAL FOR DETERMINING ALERT TRIGGER CONDITIONS.
- VALUE UPDATES AUTOMATICALLY WHEN USAGE ENTRIES ARE RECORDED.

Data Store Description Form

ID:	D1			
NAME:	INVENTORY RECORDS			
ALIAS:	STOCK MASTER FILE			
DESCRIPTION				
HOLDS CURRENT QUANTITY, ITEM DETAILS, AND THRESHOLD VALUES FOR ALL STOCK ITEMS				
DATA STORE CHARACTERISTICS				
FILE TYPE	☒ Computer	☐ Manual		
FILE FORMAT	☒ Database	☐ Indexed	☐ Sequential	☐ Direct
RECORD SIZE(CHARACTER) :		180	BLOCK SIZE: 4096	
MAX NUMBER OF RECORDS:		10000	AVERAGE: 7000	
GROWTH:		4%		
DATA SET NAME	INV_MASTER			
COPY MEMBER	INV_REC_COPY			
DATA STRUCTURE	INVENTORY RECORD			
PRIMARY KEY	ITEM ID			
SECONDARY KEYS	ITEM NAME, CATEGORY			
COMMENTS				
SOME ITEMS REQUIRE DYNAMIC THRESHOLDS DEPENDING ON SEASON OR EVENTS, WHICH COMPLICATES STATIC THRESHOLD VALUES.				

Conclusion

Ambiguities

- Unclear frequency for some low-stock alerts (real-time vs. scheduled checks).
- Exact integration scope with the IUT central database not fully defined.
- Level of detail required for archived usage logs not specified by stakeholders.

Challenges

- Mapping informal staff terminology for items to standardized database fields.
- Designing validation rules that balance strict accuracy with ease of use.
- Estimating growth rates for certain data stores without historical records.

Interesting Discoveries

- Many inventory discrepancies were traced back to manual miscounting, which the system could reduce.
- Usage logs could be mined for seasonal consumption patterns, enabling better stock planning.
- Stakeholders showed unexpected interest in having a mobile view for quick stock checks.

Thanks!

Do you have any questions?
Feel free to ask now

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